CAPSTONE DEFENCE

Certificates a shipowner can verify on their own

Fund a bounded local pilot of sovereign evidence cells.
Do not fund Azure deployment yet.

Green-Fuel Bunkering Certification Platform

Phase 0: design, controls and evidence. No application and no Azure resource exists.



SAMPLE · NOT APPROVED FOR PRESENTATION

SAMPLE · NOT APPROVED FOR PRESENTATION

A green-fuel premium is only paid if the certificate is believed

Each delivery must become a certificate that survives a shipowner, an auditor and a dispute.

1 **Meter readings** → 2 **Blend and feedstock** → 3 **Emissions attribute** → 4 **Certificate** → 5 **Shipowner and auditor**

WHERE IT BREAKS

- Evidence sits in many places
- Methods change over time
- Disputes need proof

Sources PROBLEMSTATEMENT.md · STAKEHOLDERS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Reliability decides how much of the premium is kept

The premium pool is the prize and trusted certificates protect it. How much is unknown: the business has supplied no figure.

$39.7M

premium revenue pool a year, year 3, base inputs

SCENARIO

$666,792

of it protected by the platform, base case

SCENARIO

0

inputs supplied by the business so far

NOT PROVIDED

Scenario figures are assumptions in a model, not facts. They show where value would come from.

REVIEW Evidence missing: Group delivery volumes, green-fuel premiums and dispute history

Sources BUSINESS-VALUE-MODEL.md · COMMERCIAL-VALUE.md

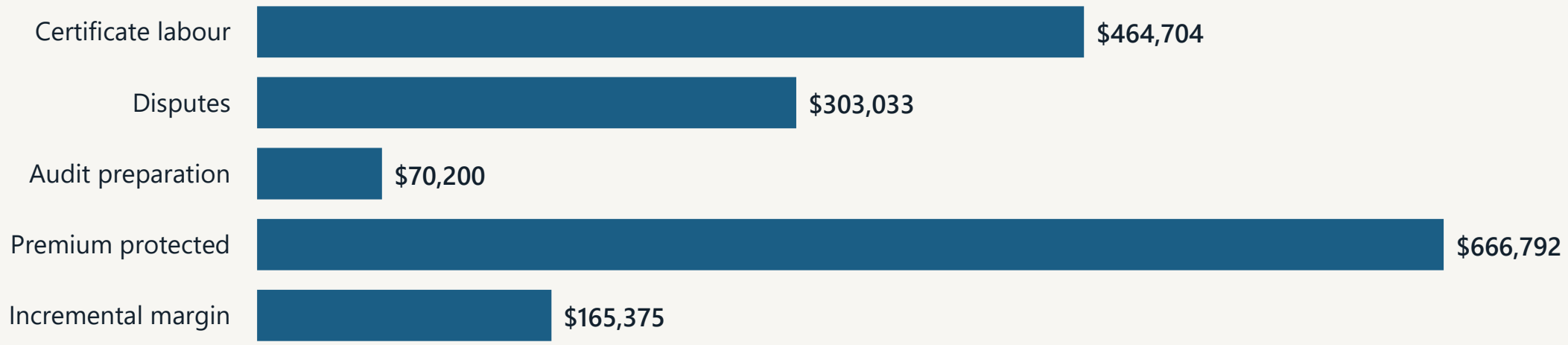
SAMPLE · NOT APPROVED FOR PRESENTATION

Base case: real value, but no three-year payback

Placeholder inputs give a negative three-year NPV (-$1.56M) and about 60 months to pay back. Volume decides it.

Annual benefit by source, year 3, base case

SCENARIO



Break-even needs about 901 deliveries a year; the base case assumes 600. Doing nothing keeps the manual cost of about $985 per certificate and forgoes about $1.7M a year.

Sources BUSINESS-VALUE-MODEL.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Today's manual process is not known well enough, and fails where it is known

Records can be altered undetectably, effort grows with volume, and audit preparation is a document hunt.

CRITERION	TODAY'S PROCESS
Data residency	where today's data sits is not known
Evidence integrity	document-based records can be altered undetectably
Scalability	effort grows with volume
Cost	no baseline cost exists (D-034)
Operational complexity	manual effort and error
Security	weak controls on paper records
Certification auditability	audit preparation is a document hunt
Vendor dependency	none, and none of the benefit

The assessment is from the options analysis. The current systems themselves are not known (D-019, D-035).

Sources 02-ARCHITECTURE-OPTIONS.md · PROBLEMSTATEMENT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Four options on eleven criteria: only sovereign cells can be made to meet residency

The cheapest option cannot show port residency. The recommended one pays for it in operational complexity.

	A CENTRALISED	B EDGE PLUS SHORE	C SOVEREIGN CELLS	D TODAY
Data residency	Does not meet	Conditional	Conditional	Unknown
Evidence integrity	Conditional	Conditional	Conditional	Does not meet
Availability	Conditional	Conditional	Conditional	Unknown
Scalability	Meets	Conditional	Conditional	Does not meet
Performance	Conditional	Meets	Meets	Unknown
Cost	Meets	Unknown	Conditional	Unknown
Operational complexity	Meets	Does not meet	Does not meet	Does not meet
Security	Conditional	Conditional	Conditional	Does not meet
Certification auditability	Conditional	Conditional	Meets	Does not meet
Recovery	Conditional	Does not meet	Conditional	Unknown
Vendor dependency	Conditional	Conditional	Conditional	Meets

Verdicts are the options analysis's own. Costs over three years: A $2.7M, B $3.8M (not like-for-like), C $5.2M.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · results.json

SAMPLE · NOT APPROVED FOR PRESENTATION

Recommend sovereign cells, a deterministic core and bounded agents

Evidence stays in its port. Software computes and signs. AI drafts and explains, and never decides.

Group plane signed methodology and policy go down; only approved-flow metadata comes up (undecided, D-011)

SOVEREIGN CELL · PORT JURISDICTION A

Edge buffer
capture, sign, queue

→

Verify and receipt
duplicates ignored

→

Evidence ledger
append-only

Deterministic services: software computes, signs, enforces
calculation under a versioned method · certificate signing · approvals · audit

Bounded agents (optional)
draft and explain; never approve

Per-shipowner access
isolation enforced by the cell

CELL · JURISDICTION B

Same contract; hosting may differ: an Azure region or a local host, decided per port.

One cell per jurisdiction, not per port.

Shipowner
verifies offline, public key only

Certification body
only after quality-manager approval

Design, not built. The group plane carries no raw or evidence-grade data.

ADR-0016 Blocked | ADR-0017 Blocked | ADR-0018 Blocked | ADR-0019 Blocked

Sources 03-ARCHITECTURE-VIEWS.md · DES-0003-target-system-architecture.md

SAMPLE · NOT APPROVED FOR PRESENTATION

It wins on the hard constraints and pays for it in complexity

Five decisive reasons, in order. Each ties to a stated requirement or constraint.

REASON	WHAT IT MEANS
Residency by construction rather than by region…	In C the unit of placement is the jurisdiction, so every store, backup, log stream…
The system of record is not at the edge	B puts the only durable copy on a small device. C treats the edge as a buffer…
Agents stay possible	SEC-003 allows AI inference over delivery data only in its jurisdiction…
Blast radius and auditability	A compromise, outage or audit touches one jurisdiction…
It degrades gracefully per port	A port with a verified in-country region runs an Azure cell…

Conditions attached: if the sponsor will not pay for sovereignty, the choice is fewer ports, not a cheaper design that breaks residency.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · 08-DESIGN-DEFENSE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Agents draft. Services validate and issue. A person approves

Five business agents work inside fixed boundaries; none can approve, file or change a method.

1
Transfer completed
signed events

→

2
Evidence bound
chain verified

→

3
Emissions attribute
deterministic code

→

4
Certificate drafted
by an agent

→

5
Approved and issued
quality manager, signed

→

6
Shipowner verifies
public key only

THE RULES THAT HOLD

Agents never hold the permission to approve or file

Emissions come from versioned methods, never from a model

Every step ends in a defined, auditable state

Sources DEMO-EXECUTIVE.md · AGENT-AUTONOMY-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Evidence stays in its port: everywhere else is conditional or prohibited

A region name is not residency proof. Placement is proposed per data class and zone, and undecided means no.

	Z1 EDGE	Z2 PORT SHORE	Z3 GROUP HUB	Z4 CLOUD/AI	Z5 CUSTOMER	Z6 STAFF ABROAD	Z7 EXTERNAL
Raw delivery measurement data	Yes	Yes	No	No	Cond	No	Cond
Delivery evidence packages	Yes	Yes	No	No	Cond	No	Cond
Blend and composition data	Yes	Yes	No	No	Cond	No	Cond
Feedstock certificates	Yes	Yes	TBD	No	TBD	No	Cond
Methodology and reference data	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calculation records	Yes	Yes	Cond	No	Cond	No	Cond
Mass-balance ledger	Yes	Yes	Cond	No	No	No	Cond
Certificates	Yes	Yes	Cond	No	Cond	No	Cond

Proposed rules. Yes permitted, Cond permitted only after a named decision, No prohibited, TBD undecided and treated as No.

ADR-0004 Blocked

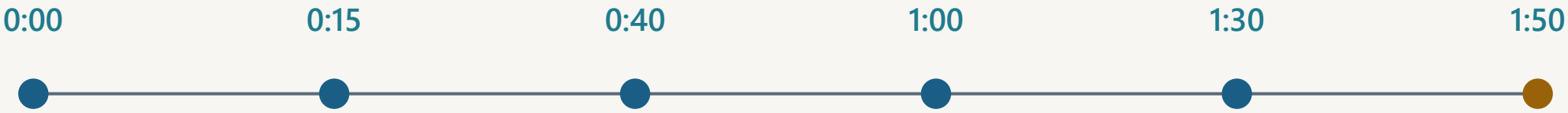
ADR-0015 Blocked

Sources SOVEREIGNTY-MATRIX.md · 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Watch a delivery become a certificate a shipowner can verify

Two minutes. Real code on synthetic data. Every step is labelled, and a recording is announced as a recording.



Transfer completed	Evidence bound	Emissions attribute	Certificate issued	Shipowner verifies	Time saved
5 signed events	chain verified, 2,500 kg	deterministic, re-run matches	human approval, signed	public key only, offline	SCENARIO, not measured

Run: node harness/demo/run.mjs exec. Backup: 1 recorded run(s), each marked "notLive".

REVIEW Evidence missing: Measured reduction in certificate processing time

Sources DEMO-EXECUTIVE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Measured: the rules do the work. AI value is unproven

On 42 synthetic cases a conventional workflow completed 41. No language model has been run.

41 of 42

synthetic cases completed, no model

MEASURED

15 of 15

emissions calculations correct

MEASURED

0

language-model arms run

NOT RUN

Synthetic, author-written, unreviewed; the holdout is not blind. No throughput, latency or availability test exists: there is no system to test.

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

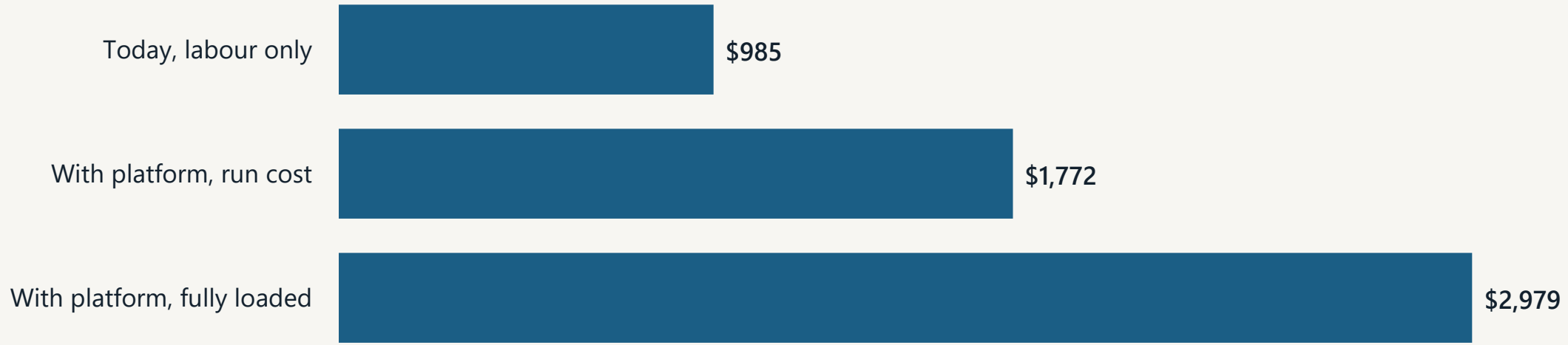
SAMPLE · NOT APPROVED FOR PRESENTATION

Base case: $5.2M over three years, and more per certificate than today

Cost per certificate rises. The case rests on disputes and premium protection, not on labour.

Cost per certificate, base case

SCENARIO



One-time $2.3M; annual operating $985,308. 12 verified inputs (Azure list prices) and 79 scenario assumptions.

Sources BUSINESS-VALUE-MODEL.md · azure-prices.json

SAMPLE · NOT APPROVED FOR PRESENTATION

Decision: fund a bounded local pilot. Not Azure deployment yet

Approve Phase 1a on synthetic data, and supply four inputs that turn assumptions into numbers.

THE ASK	WHAT IT UNLOCKS
Approve Phase 1a: local, synthetic, no Azure. Enrol an approver	The first vertical slice
Real volumes, premiums and dispute history	A business case built on facts
Name the ports. Counsel's reading on residency	A hosting model per port
A domain reviewer. Decide whether to test an LLM	A reviewed evaluation

Revisit the whole case if confirmed volume is well below about 901 deliveries a year.

Sources PHASE-1-BLOCKER-REPORT.md · TODO.md

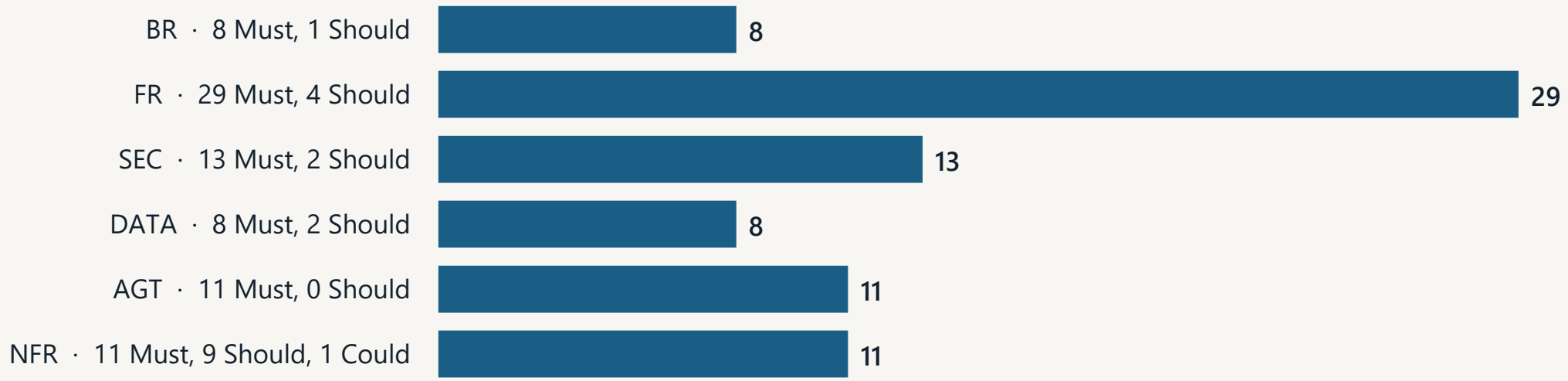
SAMPLE · NOT APPROVED FOR PRESENTATION

99 requirements, 80 of them Must, in six families

Case-study requirements are kept apart from assumptions, open questions and claims that still need verification.

Must requirements by family

CASE STUDY



Counts are read from the requirements baseline; a Must requirement without a decision is listed in the appendix.

Sources REQUIREMENTS.md · REQUIREMENTS-TRACEABILITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Azure services are provisional leans, each with a residency check still to run

Selected by requirement, not by name. Alternatives were considered, and the open question per service is stated.

ROLE	CANDIDATE	DISPOSITION
Services and agent runtime	Azure Container Apps	Provisional lean
Ledger and metadata	Azure Database for PostgreSQL	Provisional lean
Evidence and certificates	Azure Blob Storage	Provisional lean
Ingestion queue	Azure Service Bus	Provisional lean
Signing keys	Azure Key Vault Managed HSM	Provisional lean
Tamper-evidence anchor	Azure Confidential Ledger	Candidate, undecided
Agent models	Microsoft Foundry model deployments (Azure OpenAI models)	Deferred
Identity	Microsoft Entra ID	Provisional lean

16 roles are assessed in the service decision matrix. None is verified in a region and none is deployed.

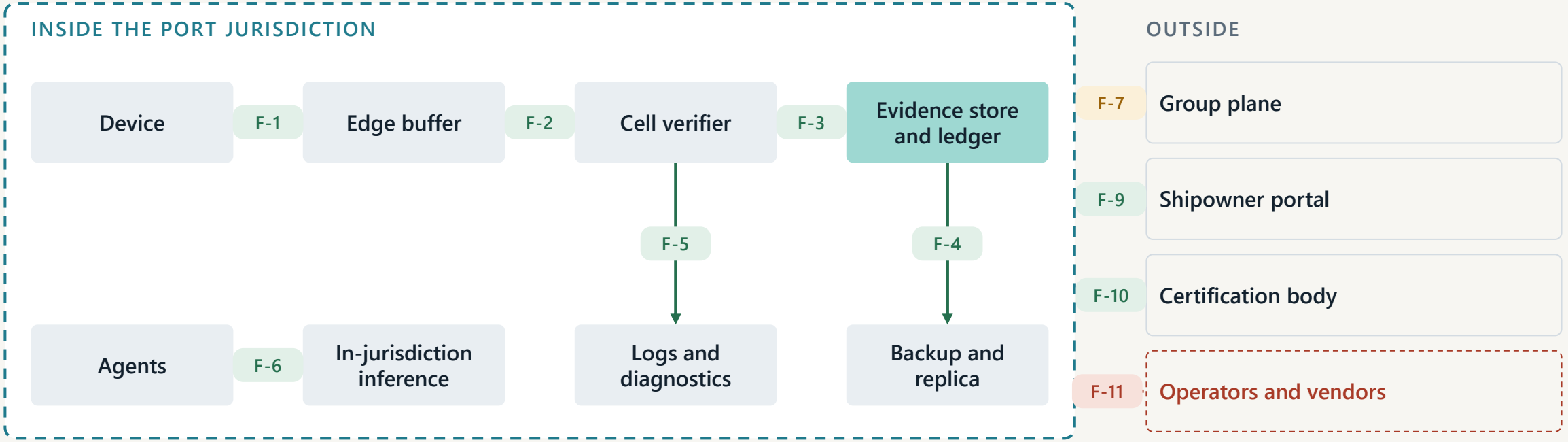
ADR-0019 Blocked

Sources 07-AZURE-SERVICE-DECISION-MATRIX.md · AZURE-REFERENCE-ARCHITECTURE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Thirteen flows are classified. Only some may leave the jurisdiction

Edge links may be intermittent: events are buffered, signed and retried idempotently. Retries are duplicates, never new evidence.



Arrow colour: green allowed inside the jurisdiction, amber undecided, red forbidden by default. All 13 flows are in the appendix register.

ADR-0004 Blocked

Sources 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md · 03-ARCHITECTURE-VIEWS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Four agents may reach L4. One stays at L3. None can approve or file

L4 is a ceiling on autonomy inside a fixed boundary, not a permission to act commercially, regulatorily or on infrastructure.

	L0	L1	L2	L3	L4	L5	WHY THIS CEILING
Delivery Evidence (DEA)							Ingests and binds source evidence as received within a fixed catalogue
Emissions Attribute (EAA)							Runs approved deterministic calculations…
Dispute Resolution (DRA)							Assembles packs; every outward or commercial step is human
Mass-Balance (MBA)							Posts events derived from signed source events…
Customer Compliance (CCA)							Answers in-policy questions directly to customers, with sampled review

L0 No AI · L1 Assistive · L2 Drafting · L3 Conditional action · L4 Bounded autonomy · L5 Unbounded autonomy

Proposed ceilings. Approval, filing, method change and infrastructure change are never agent actions.

ADR-0017 Blocked

Sources AGENT-AUTONOMY-MATRIX.md · AZURE-REFERENCE-ARCHITECTURE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

59 controls in seven families. 9 cannot be tested until a product exists

Shipowner isolation is enforced per tenant by the cell, not by the agent. Each control names the requirement it serves.

Controls by family

CANDIDATE



Security review: three HIGH, eleven MEDIUM and two LOW. Candidate oracles test the rules on synthetic data; no control is implemented in a product.

ADR-0017 Blocked

Sources 10-CONTROL-MATRIX.md · 13-SECURITY-REVIEW.md · 06-THREAT-MODEL.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Evidence is a signed chain; the certificate binds to its hash

A verifier needs only the certificate, the evidence files and a trust set it supplies. Tampering is detected, not prevented.

1
Device signs each event
key held by the device

2
Cell verifies and receipts
signed receipt, own clock

3
Events chain by hash
each hashes the previous

4
Manifest binds evidence
one hash for the transfer

5
Certificate binds manifest
quality manager signs

6
Shipowner verifies
public key, offline

Tampering is detected, not prevented. Change any event, receipt, manifest or certificate and a signature or hash no longer matches. An independent external anchor is an open decision (ADR 0018).

Recorded run (not live): 5 signed events, chain verified, 2,500 kg delivered, 5 receipts, tamper attempt detected.

ADR-0018 Blocked

Sources custody-chain-contract.mjs · certificate-lifecycle-contract.mjs · tamper-20261029T131741-5b3008eb.json

SAMPLE · NOT APPROVED FOR PRESENTATION

Scalability is argued, not measured

The argument is structural: each cell scales independently. No throughput, latency or scale test has been run, because there is no system to test.

REVIEW PLACEHOLDER

EVIDENCE NEEDED

Measured throughput, latency and scaling under load, per cell.

WHAT THE REPOSITORY HOLDS INSTEAD

- Placeholders and scenario arithmetic only; no target volume or response time exists (D-031, NFR-014, NFR-015).
- PH-C1: Deliveries per cell per year
- PH-C2: Peak concurrent deliveries per port
- PH-C3: Meters per delivery and reading rate
- PH-C4: Ingestion rate at peak

REVIEW Evidence missing: Measured throughput, latency and scaling evidence

Sources 05-QUALITY-ATTRIBUTES.md · COST-AND-CAPACITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Recovery stays inside each jurisdiction. Targets are unset and untested

Under failure the system refuses, waits or escalates. It never guesses. Regional outage and recovery are designed, not tested.

LAYER	BASELINE DESIGN	STILL OPEN
Evidence blobs	Locally or zone redundant only…	Whether a second region in the same…
Ledger database	High availability inside the region as…	Recovery targets (D-031)…
Messaging	Zone redundancy where the tier supports it…	Replay rules after a failure…
Keys and certificates	Key custody in the jurisdiction (HC-03)…	HSM choice and ownership (D-013)
Edge	Store-and-forward with a local append-only…	Maximum disconnection and permitted…
Control plane and IaC	The repository and reviewed templates…	Where the pipeline runs and what it can see
Customer-visible service	Customer compliance answers can be degraded…	The degraded-mode behaviour per agent

Edge outage, tamper, replay and injection are tested on candidate contracts. Region outage is not.

REVIEW Evidence missing: Tested recovery against agreed RPO and RTO

Sources OPERATIONS-AND-DR.md · 08-DESIGN-DEFENSE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Observability is designed per cell. Nothing is instrumented yet

Logs and metrics stay with the data they describe. Telemetry is classified so it cannot leak what the cell must not export.

NEED	DESIGN	NOT YET DECIDED OR VERIFIED
Logs and metrics stay with the…	One workspace per jurisdiction cell…	That the chosen region meets D-009 for logs…
No sensitive data in telemetry	Telemetry is classified; names of customers or vessels…	A logging standard for the application…
Application and dependency health	Application Insights workspace-based…	Instrumentation: no application exists
Evidence-pipeline health	Alerts on ingestion lag, dead-letter growth…	Thresholds: no target is agreed (D-031)
Edge health	Edge reports its last synchronisation…	The edge platform (D-018)
Audit	Every approval, every agent action and every denial is…	The audit store design

A recorded observability demonstration exists (not live). No application emits telemetry.

REVIEW Evidence missing: Operational traces, dashboards and alert history from a running system

Sources OPERATIONS-AND-DR.md · observability-20261029T131742-b48a4268.json

SAMPLE · NOT APPROVED FOR PRESENTATION

Four arms were scored. Both language-model arms are not run

The decision rule is fixed first: use rules by default and admit a model only where it beats the measured baseline.

ARM	RESULT	STATE
Conventional workflow (no model)	41 of 42 complete; all 9 hard gates pass	MEASURED
Four deliberately bad arms	All fail the gates, as they must (scorer check)	MEASURED
Single-agent language model	No run	NOT RUN
Multi-agent language model	No run	NOT RUN

Synthetic, author-written, unreviewed cases; the holdout is not blind. Model selection is deferred (D-028).

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

What was rejected, and the requirement that rejected it

Every rejection is stated in terms of a requirement or constraint, not preference.

ALTERNATIVE	DISPOSITION	REASON
A Centralised Azure	Rejected (hard constraint)	Violates HC-01 for any group with more than one port jurisdiction…
B Edge plus shore	Not selected	Not a hard-constraint violation if the shore platform is confined to derived data, but it…
D Today	Not selected	Does not meet HC-08 (undetectable alteration), does not scale with volume…
Azure Functions	Not selected	Adds a second programming and operations model for little gain at this volume…
App Service	Not selected	No requirement that Container Apps cannot meet; one fewer service to verify

What would change the recommendation is listed in the options analysis.

ADR-0016 Blocked

Sources 02-ARCHITECTURE-OPTIONS.md · 07-AZURE-SERVICE-DECISION-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

6 critical gaps stand between this repository and a defensible capstone

They are stated before anyone asks. Each says how it is closed and by whom.

GAP	WHAT IS MISSING	CLOSED BY
G-01	No working solution end to end. No src/, no business agent, no Azure resource…	build, owner
G-02	No AI solution and no model evaluation of any business agent…	build
G-03	No numbers the audience can act on. No latency, throughput, cost per delivery…	build, owner
G-04	No decided solution architecture and no recommendation…	owner, build
G-05	Thought-leadership artifact not chosen, finished or submitted…	owner, publish
G-06	No narrative, deck, 2-minute demo or rehearsal material. Section B…	build, live

From the forensic audit at the commit it names; gaps closed since are not re-marked here.

Sources CRITICAL-GAPS.md · RUBRIC-TRACEABILITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

The path to production runs through nine gates. None is passed

Production deployment is never automatic: each gate needs a person, and Azure deployment is deferred by the owner.



Gate G1 (phase raised) and G2 (approver enrolled) come first. Architecture baseline authorization is built; no baseline exists.

Sources PHASE-1-BLOCKER-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

The target architecture is Blocked in the register, not approved

Status is read from the decision register at this commit, never written into the deck.

ADR	DECISION	STATUS	APPROVAL BASIS
ADR-0004	Hold delivery measurements in jurisdiction-local authoritative stores	Blocked	none
ADR-0015	Jurisdiction-local, tamper-evident evidence ingestion by an edge…	Blocked	none
ADR-0016	Sovereign evidence cells per jurisdiction with federated coordination	Blocked	none
ADR-0017	Deterministic core and bounded business agents	Blocked	none
ADR-0018	Evidence integrity and certificate mechanism…	Blocked	none
ADR-0019	Azure services are selected by requirement and verified per…	Blocked	none
ADR-0020	Phase 1a, a local synthetic-data vertical slice inside this repository	Proposed	none

12 of 57 register decisions are approved; 45 are not yet decided.

ADR-0004 Blocked ADR-0015 Blocked ADR-0016 Blocked ADR-0017 Blocked

Sources 0016-sovereign-evidence-cells-with-federated-coordination.md · 0020-phase-1a-local-synthetic-vertical-slice.md

SAMPLE · NOT APPROVED FOR PRESENTATION

3 Must requirements have no decision yet

Each requirement traces to the decisions it depends on, with its acceptance criteria and evidence required.

FAMILY	REQUIREMENTS	MUST	TRACED TO A DECISION
BR	9	8	8
FR	33	29	33
SEC	15	13	14
DATA	10	8	8
AGT	11	11	9
NFR	21	11	19

Must requirements without a decision: BR-009, DATA-003, AGT-009.

Sources REQUIREMENTS-TRACEABILITY.md · REQUIREMENTS-COVERAGE.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Reference topology: one Azure cell per port jurisdiction, design only

No Azure resource has been created. The icons are placeholders until the official icon set is supplied.

Edge buffer

Arc or IoT candidate; platform undecided (D-018)

Group plane

signed policy down; approved flows up

ONE AZURE CELL PER PORT JURISDICTION · DESIGN ONLY, NOTHING DEPLOYED

EN **Microsoft Entra ID** human and workload identity	GW **Container Apps ingress** API gateway: APIM candidate	PL **Azure Policy and RBAC** location and access guardrails
CA **Azure Container Apps** services and agent runtime	SB **Azure Service Bus** ordered, deduplicated intake	AI **Foundry model** optional, in-jurisdiction only
PG **Azure PostgreSQL** ledger and mass balance	BL **Blob Storage, immutable** evidence and certificates	KV **Key Vault Managed HSM** keys held in the jurisdiction
MO **Azure Monitor** one workspace per cell	CL **Confidential Ledger** anchor undecided (ADR 0018)	CR **Container Registry** signed images, promotion

Provisional leans from the service decision matrix. Region, quota and residency are unverified for every service.

ADR-0019 Blocked

Sources AZURE-REFERENCE-ARCHITECTURE.md · 07-AZURE-SERVICE-DECISION-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

The 13 classified data flows, with the design position of each

Only some may leave the jurisdiction. Validation of each flow against a real network path is still to do.

FLOW	FROM TO	DESIGN POSITION
F-1	Device to edge buffer	Allowed, inside the jurisdiction, store-and-forward
F-2	Edge buffer to cell verifier	Allowed, inside the jurisdiction
F-3	Cell verifier to cell evidence store and ledger	Allowed, inside the jurisdiction
F-4	Cell to cell backup and replica	Allowed only to a target inside the jurisdiction
F-5	Cell to cell log and diagnostic store	Allowed, inside the jurisdiction; export only on an approved flow
F-6	Cell agents to the inference endpoint	Allowed only to an endpoint in the jurisdiction whose processing location is…
F-7	Cell to group plane	Undecided. Each item needs a class and an approved-flow entry with a legal basis
F-8	Group plane to cell	Allowed; signed, verified at the cell
F-9	Cell to shipowner portal	Allowed; per-tenant
F-10	Cell to certification body	Allowed only after quality-manager approval, via the controlled submitter
F-11	Any operator or vendor path into a cell from outside	Forbidden by default; break-glass approved inside the jurisdiction
F-12	Deployment pipeline to a cell	Allowed through scoped deployment identities; must not be a way to read cell data
F-13	Real data to any non-production environment	Forbidden; synthetic or approved-masked data only

ADR-0004 Blocked

Sources 12-DATA-MOVEMENT-AND-RESIDENCY-VALIDATION.md

SAMPLE · NOT APPROVED FOR PRESENTATION

There is no performance evidence yet

This slide is a placeholder by design: it is replaced when a running slice is measured.

REVIEW PLACEHOLDER

EVIDENCE NEEDED

Throughput, latency, scale and availability measurements with method, environment, and raw results.

WHAT THE REPOSITORY HOLDS INSTEAD

- No application and no Azure resource exist.
- Capacity method exists; every figure is withheld (see cost and capacity).

REVIEW Evidence missing: Performance results

Sources COST-AND-CAPACITY.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Cost: the method exists; the Azure run cost has no figure

79 scenario assumptions, 12 verified inputs, 1 benchmark, 0 supplied by the business.

ITEM	WHY THERE IS NO FIGURE
Monthly run cost per jurisdiction cell	No SKU, volume, retention, redundancy or region is decided
Cost of the edge or on-premises evidence store	The edge platform and hardware ownership are open (D-018, D-010)
Cost of model use	The models, deployment types and prompt volumes are open (D-028)
Non-production cost	The number and scale of environments is open (D-037)
Budget	The cost envelope is open (D-034)
Capacity and quotas	No target volume or response time exists (D-031, NFR-014, NFR-015)…

The business value model uses Azure list prices only as verified inputs; its totals are scenario results.

Sources COST-AND-CAPACITY.md · BUSINESS-VALUE-MODEL.md

34 threats across six trust boundaries, each with a control and a stated limit

The security review found three HIGH findings; none is closed by a product because none exists.

34

threats in the threat model

DESIGN

59

controls mapped

CANDIDATE

9

controls not yet testable

NOT BUILT

Security review: three HIGH, eleven MEDIUM and two LOW. Boundaries: device to edge, edge to cell, agents to tools, shipowner to portal, group plane to cell, certificate to outside.

Sources 06-THREAT-MODEL.md · 13-SECURITY-REVIEW.md · 10-CONTROL-MATRIX.md

SAMPLE · NOT APPROVED FOR PRESENTATION

41 of 42 synthetic cases complete; the one miss is a free-text relative date

Hard gates passed. Language-model arms did not run.

97.6%

task completion rate

MEASURED

15 of 15

emissions calculations correct

MEASURED

9 of 9

hard gates passed

MEASURED

Run AR-deterministic-workflow-3f7a0688. Synthetic cases written by the author and not reviewed; the holdout is not blind.

Sources COMPARISON.md · AGENT-EVALUATION-REPORT.md

SAMPLE · NOT APPROVED FOR PRESENTATION

1,056 of 1,056 automated tests pass on the recorded run

They test the harness and candidate contracts on synthetic data. They do not show a requirement is satisfied.

1,056

tests passing, recorded run EV-0077

MEASURED

0

application tests

NOT BUILT

0

end-to-end tests of the platform

NOT BUILT

Requirement coverage is generated and states what a test that ran has shown.

Sources EV-0077.json · REQUIREMENTS-COVERAGE.md

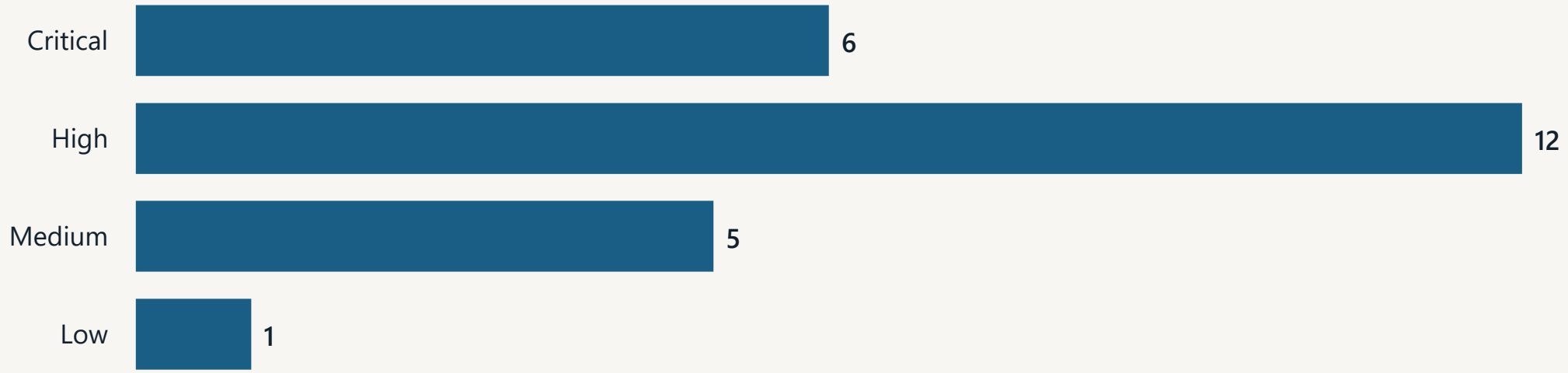
SAMPLE · NOT APPROVED FOR PRESENTATION

24 gaps were recorded at the audit. 6 are critical

The register is dated to its audit commit. A gap closed since is not marked closed here.

Gaps by severity at the audit

DESIGN



Rubric floors: no criterion scored 1; each section at least 60 percent; artifact submitted.

Sources CRITICAL-GAPS.md

SAMPLE · NOT APPROVED FOR PRESENTATION

Fifty panel questions are prepared. One per area is shown here

Each has a recommended answer, its evidence, what is uncertain, and when to concede.

ID	AREA	QUESTION
Q01	Discovery	Have you actually spoken to a bunker supplier, or is this a desk exercise?
Q02	Alternatives	Why not use spreadsheets, an ERP module or an existing certification registry?
Q03	Commercial	Where does the premium revenue figure come from?
Q14	Architecture	One cell per port looks over-engineered. Defend it.
Q15	AI governance	The brief says L4 agents. What stops an agent doing harm?
Q18	Delivery	You built a harness and no product. Is this over-engineered for a capstone?
Q21	Demo	Was that live or recorded?
Q24	Disaster recovery	What happens if the Azure region goes down?
Q26	Security	What if a signing key is stolen?

Full answers: docs/capstone/presentation/QA-PANEL-50.md. Delivery under challenge is not verified until a human rehearsal.

Sources QA-PANEL-50.md

Presentation criteria: what the deck can serve and what only a live session can

No official score is assigned. Delivery, listening and composure are not verified until a person has rehearsed.

CRITERION	EVIDENCE NOW	SLIDES	HOW ASSESSED
B1 Narrative structure & argument	None	E01, E02, E05, E06	L, H
B2 Audience calibration & register	Thin	T02, T03, T04, T05	L, H
B3 Demo & executive hook	Thin	E09, E11	L
B4 Q&A composure & depth under…	Partial	T02, T03, T04, T05	L, H
B5 Delivery craft: pace…	None	live only	L
B6 Written communication…	Partial	E07, T01, A01, A02	T, H
B7 Use of evidence & data…	Thin	E03, E04, E06, E08	T, L
C1 Discovery & business framing	Documented	E02, E03, E04, E05	L, H
C2 Active listening & disagreement…	None	A10	L
C3 Influence & intellectual honesty	Demonstrated honesty	E01, E03, E04, E12	L, H

Mode: T technical evidence, L scored live, H human judgment, P publication.

REVIEW Evidence missing: Human rehearsal of the B and C criteria

Sources RUBRIC-TRACEABILITY.md · REHEARSAL-PLAN-AND-SCORECARD.md

This deck was generated from one commit and a recorded set of sources

A slide is only as good as the evidence it cites. The generation record beside the file lists every source with its hash.

ITEM	VALUE
Repository commit	7faeb8c42df4e2a57adbc3005e6e2d4e386afce2
Branch	studio/ui-readonly
Slides	40
Sources read	35 files, digest cce969bda65a134f
Claims checked	39; 0 unsupported
Rubric	derived traceability; the official PDF is not in the repository